Report Date:January 23, 2018 File:120

Report Number:076276

Town of Ladysmith

PO Box 220, 410 Esplanade
Ladysmith BC V9G 1A2

Dear Town of Ladysmith

Re: Non-compliance Advisory Letter, Permit 120, Ladysmith, BC, Effluent

On January 11, 2018, Ministry of Environment and Climate Change Strategy, Environmental Protection Division staff conducted an inspection of your facility, Town of Ladysmith municipal sewage system located at 245 Oyster Cove Road, Ladysmith, BC with authorization number 120 under the *Environmental Management Act*. Ministry staff were accompanied on site by Curtis Baker, Town of Ladysmith; Geoff Goodall, Town of Ladysmith.

This Advisory, the alleged violations and the circumstances to which it refers will form part of the compliance history of Town of Ladysmith, and will be taken into account in the event of future non-compliance.

Please note that this authorization is considered to be out of compliance until such a time as it can be confirmed to meet the authorization requirements.

Inspection Details:

Requirement Description:	**1. AUTHORIZED DISCHARGES, 1.1** 1.1.1: The discharge of effluent to which this Section is applicable is from a municipal sewage system as shown on the attached Site Plan A. The reference number (S.E.A.M. site number) for this discharge is E100004. 1.1.1 The maximum rate at which effluent may be discharged through the sewage treatment plant is 6 100 m3/d; the annual average rate of discharge is 3 050 m3/d. Flow in excess of treatment plant capacity is to be discharged to the bypass outfall. The maximum rate at which effluent may be bypassed is 8 500 m3/d; the total annual volume which may be discharged to the bypass outfall is 228 000 m3.
Details/Findings:	On January 11, 2018 BC Ministry of Environment and Climate Change Strategy ("ENV") Environmental Protection Officer Darren Stewart ("Officer Stewart") inspected the Town of Ladysmith ("ToL") municipal sewage system for compliance with its effluent discharge permit PE-120. A data review was also performed by Officer Stewart for date range April 2016 thru December 2017 ("Q2 2016 - Q4 2017"). Officer Stewart observed 12 exceedances of the maximum rate of discharge in data submitted by ToL: October 14, 2016 = 9758 m3/d, October 15, 2016 = 7836 m3/d, November 02, 2016 = 7808 m3/d, November 26. 2016 = 7301 m3/d, December 20, 2016 = 6103 m3/d, January 18, 2017 = 7472 m3/d, February 15, 2017 = 11332 m3/d, February 16, 2017 = 7511 m3/d, November 14, 2017 = 7427 m3/d, November 15. 2017 = 9125 m3/d, November 21, 2017 = 6208 m3/d, and November 22, 2017 = 6527 m3/d. The annual average rates of discharge for 2016 (2725 m3/d) and 2017 (2768 m3/d) were both compliant with this section.

Ministry of Environment and Climate Change Strategy

Compliance
Environmental Protection Division

Mailing Address:
2080-A
Labieux Rd
Nanaimo BC V9E 6J9

Telephone: 250 751 3100
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Compliance:	Out
Actions to be taken:	
Requirement Description:	**1. AUTHORIZED DISCHARGES, 1.1** 1.1.2: The discharge of effluent to which this Section is applicable is from a municipal sewage system as shown on the attached Site Plan A. The reference number (S.E.A.M. site number) for this discharge is E100004. 1.1.2 The characteristics of the discharge shall not exceed: Sewage Treatment Plant Discharge: 5-day Biochemical Oxygen Demand: 130 mg/L. Total Suspended Solids: 130 mg/L. Overflow Bypass Discharge: Screened combined stormwater and sewage effluent.
Details/Findings:	Officer Stewart observed 2 discharge characteristic exceedances in data submitted by ToL: April 05, 2016, 5-day Biochemical Oxygen Demand = 155 mg/L June 09, 2016, 5-day Biochemical Oxygen Demand = 138 mg/L
Compliance:	Out
Actions to be taken:	
Requirement Description:	**1. AUTHORIZED DISCHARGES, 1.1** 1.1.3: The discharge of effluent to which this Section is applicable is from a municipal sewage system as shown on the attached Site Plan A. The reference number (S.E.A.M. site number) for this discharge is E100004. 1.1.3 The works authorized are influent screening, a Spiragester primary type treatment plant, chlorination and dechlorination facilities, autothermophilic aerobic sludge processing facilities, foul air scrubber, sludge dewatering facilities, overflow bypass, bypass screening and solids removal facilities, 300 mm and 710 mm diameter parallel outfalls extending 850 m from mean low water to a depth of 17.9 m below mean low water, and related appurtenances approximately located as shown on the attached Site Plan A.

Details/Findings:	Mr. Goodall informed Officer Stewart that a new treatment plant consisting of a moving bed bioreactor ("MMBR") was constructed and operational as of June 2016, thus replacing the Spiragester primary type treatment plant. Officer Stewart observed the MMBR in operation during the inspection although it is not an authorized work as specified by this section. Due to changes in Ministry guidance and direction for the Town of Ladysmith regarding an updated authorization either via a registration under the Municipal Wastewater Regulation ("MWR"), an Operational Certificate under a Liquid Waste management Plan ("LWMP") or a permit amendment, compliance with this requirement was not determined.
Compliance:	Not Determined
Actions to be taken:	Follow up with the BC Ministry of Environment and Climate Change Strategy, South Authorizations section which will provide guidance and recommendations for the appropriate form of authorization for this facility. Please contact the South Authorizations Section Head, Bryan.Vroom@gov.bc.ca for further information.
Requirement Description:	**1. AUTHORIZED DISCHARGES, 1.1** 1.1.6: The discharge of effluent to which this Section is applicable is from a municipal sewage system as shown on the attached Site Plan A. The reference number (S.E.A.M. site number) for this discharge is E100004. 1.1.6 The location of the point of discharge is Ladysmith Harbour.
Details/Findings:	Mr. Goodall confirmed the location of the point of discharge is Ladysmith Harbour, more specifically 48° 59.00’ north latitude and 123° 47.46’ west longitude, as specified in the facility Conditional Management Plan.
Compliance:	In
Actions to be taken:	
Requirement Description:	**2. GENERAL REQUIREMENTS, 2.1 Maintenance of Works and Emergency Procedures** 2.1: The Permittee shall inspect the pollution control works regularly and maintain them in good working order. In the event of an emergency or condition beyond the control of the Permittee which prevents continuing operation of the approved method of pollution control, the Permittee shall immediately notify the Regional Waste Manager and take appropriate remedial action.

Details/Findings:	Mr. Baker confirmed with Officer Stewart that facility staff regularly inspect the pollution control works and fill out daily checklists during their walkthrough of the treatment plant.
Compliance:	In
Actions to be taken:	
Requirement Description:	**2. GENERAL REQUIREMENTS, 2.2 Bypasses** 2.2: The discharge of effluent which has bypassed the designated treatment works is prohibited unless the consent of the Regional Waste Manager is obtained and confirmed in writing.
Details/Findings:	Mr. Baker confirmed with Officer Stewart that there have been no bypasses the designated treatment works during the inspection period, therefore compliance with this requirement was not applicable.
Compliance:	Not Applicable
Actions to be taken:	
Requirement Description:	**2. GENERAL REQUIREMENTS, 2.3 Process Modifications** 2.3: The Permittee shall notify the Regional Waste Manager prior to implementing changes to any process that may affect the quality and/or quantity of the discharges.
Details/Findings:	ToL has notified ENV on multiple occasions and in numerous methods (in-person meetings, written and phone call) regarding the upgrading of the works to include a MMBR that would affect the quality of the discharge.
Compliance:	In

Actions to be taken:	
Requirement Description:	**2. GENERAL REQUIREMENTS, 2.4 Posting of Outfalls** 2.4: The Permittee shall erect a sign along the alignment of the outfalls above high water mark. The sign shall identify the nature of the works. The wording and size of the sign requires the consent of the Regional Waste Manager.
Details/Findings:	Officer Stewart observed a sign in the waters of Ladysmith Harbour that Mr. Baker confirmed was the outfall sign for the wastewater treatment facility. However, due to access issues Officer Stewart did not assess the shoreline for a sign erected along the alignment of the outfall above the high water mark, therefore compliance with this requirement was not determined.
Compliance:	Not Determined
Actions to be taken:	
Requirement Description:	**2. GENERAL REQUIREMENTS, 2.5 Outfall Inspection** 2.5: The Permittee shall conduct a dye test on the outfall lines (or inspect by another method approved by the Regional Waste Manager) every five years or as may otherwise be required by the Regional Waste Manager.
Details/Findings:	The last outfall inspection and report performed by Sealegs Marine Services on file with ENV is dated September 13, 2011 which is in excess of the five year time frame specified by this permit section.
Compliance:	Out
Actions to be taken:	Perform an outfall inspection as required by this section and submit the resulting outfall inspection report to ENV.

Requirement Description:	**2. GENERAL REQUIREMENTS, 2.6 Chlorination** 2.6: Maintain a chlorine residual (at the point of discharge or prior to dechlorination) between 0.1 and 1.0 mg/L at all times.
Details/Findings:	The chlorine residual analytical results reported in the 2016 quarterly reports indicate that the chlorine residual was below 0.1 mg/L on six sampling days in 2016: April 24, 2016 = 0.08 mg/L, July 09, 2016 = 0.09 mg/L, August 13, 2016 = 0.05 mg/L, August 14, 2016 = 0.08 mg/L, August 16, 2016 = 0.09 mg/L, and August 17, 2016 = 0.04 mg/L. The chlorine residual analytical results reported in the 2016 quarterly reports indicate that the chlorine residual was above 1.0 mg/L on two sampling days in 2016: June 25, 2016 = 1.01 mg/L and September 04, 2016 = 1.13 mg/L
Compliance:	Out
Actions to be taken:	
Requirement Description:	**2. GENERAL REQUIREMENTS, 2.7 Dechlorination** 2.7: The effluent shall be dechlorinated prior to discharge to reduce the chlorine residual below detectable limits.
Details/Findings:	Mr. Baker confirmed with Officer Stewart that the effluent is dechlorinated prior to discharge using Captor.
Compliance:	In
Actions to be taken:	
Requirement Description:	**2. GENERAL REQUIREMENTS, 2.8 Sludge Use and Disposal** 2.8: The use and/or disposal of sludge from the treatment facilities shall be in a manner acceptable to the Regional Waste Manager.

Details/Findings:	Mr. Goodall informed Officer Stewart that liquid sludge was currently being sent approximately three times per week to the Catalyst Crofton pulp mill for treatment in their UNOX effluent plant. Dewatered sludge was also being sent to a treatment facility in Cumberland. Mr. Goodall confirmed with Officer Stewart that by the end of January 2018 that all of sludge from the ToL wastewater system will be sent to Ladysmith's own site registered under the Organic Matter Recycling Regulation ("OMRR").
Compliance:	In
Actions to be taken:	
Requirement Description:	**2. GENERAL REQUIREMENTS, 2.9 Odour Control** 2.9: Should odours from the works authorized in Section 1.1.3 become objectionable, the Permittee shall be required to provide additional works.
Details/Findings:	Officer Stewart did not observe objectionable odours from the works during the site inspection, therefore compliance with this requirement was not applicable.
Compliance:	Not Applicable
Actions to be taken:	
Requirement Description:	**2. GENERAL REQUIREMENTS, 2.11 Facility Classification** 2.11: The Permittee shall classify the wastewater treatment facility authorized in Section 1.1.3 (the facility) and the classification shall be maintained with the "British Columbia Water and Wastewater Operators Certification Program Society" (BCWWOCPS).
Details/Findings:	The Town of Ladysmith wastewater treatment plant authorized under permit 120 is currently classified as a MWWT-III (facility # 65) by the Environmental Operators Certification Program ("EOCP").

Compliance:	In
Actions to be taken:	
Requirement Description:	**3. MONITORING AND REPORTING REQUIREMENTS, 3.1 Discharge Monitoring, 3.1.1 Sampling and Analysis** 3.1.1: The Permittee shall install a suitable sampling facility and obtain a composite sample of the effluent from the treatment plant effluent once every month. The sample is to consist of four grab samples taken over a two-hour period at maximum flow and mixed to form a single sample for subsequent analysis. Proper care should be taken in sampling, storing and transporting the sample to adequately control temperature and avoid contamination, breakage, etc. Obtain analyses of the sample for the following: 5-day Biochemical Oxygen Demand; Total Suspended Solids.
Details/Findings:	Officer Stewart observed a suitable sampling facility and composite sampler able to obtain the necessary samples specified in this section. A records review showed analyses performed of the samples for 5-day Biochemical Oxygen Demand and Total Suspended Solids as required, in addition to fecal coliforms.
Compliance:	In
Actions to be taken:	
Requirement Description:	**3. MONITORING AND REPORTING REQUIREMENTS, 3.1 Discharge Monitoring, 3.1.2 Flow Measurement** 3.1.2: Provide and maintain suitable flow measuring devices and record once per day the effluent volume discharged over a 24-hour period from the sewage treatment plant and the overflow bypass.
Details/Findings:	Officer Stewart observed a magmeter maintained as a suitable flow measuring device, which records once per day the effluent volume discharged over a 24-hour period.
Compliance:	In

Actions to be taken:	
Requirement Description:	**3. MONITORING AND REPORTING REQUIREMENTS, 3.2 Sludge Monitoring** 3.2: The Permittee shall obtain a representative sample of the treated sludge once per year and obtain analyses of the sample for the following: Metals: Arsenic, Boron, Cadmium, Cobalt, Lead, Mercury, Molybdenum, Nickel, Zinc. Salmonella, Fecal Coliform, Total Kjeldahl Nitrogen.
Details/Findings:	The sludge analysis report for 2016 was submitted April 04, 2016 and the 2017 sludge report was submitted January 12, 2018. The submitted reports included analyses of the sample for fecal coliforms, CSR/CCME metals, moisture, pH, salmonella and Total Kjeldahl Nitrogen as required.
Compliance:	In
Actions to be taken:	
Requirement Description:	**3. MONITORING AND REPORTING REQUIREMENTS, 3.4 Reporting** 3.4: Maintain data of analyses and flow measurements for inspection and every three months submit the data, suitably tabulated, to the Regional Waste Manager for the previous quarter. The first report is to be submitted by January 31, 1994. Based on the results of the monitoring program, the Permittee monitoring requirements may be extended or altered by the Regional Waste Manager.
Details/Findings:	Officer Stewart observed data of analyses and flow measurements maintained for inspection, and quarterly reports are submitted as required to the ENV Authorizations Reporting inbox as required.
Compliance:	In
Actions to be taken:	

Compliance History:
April 19, 2017 - IR49542 and Advisory for insufficient chief operator requirements (this IR was not a full inspection, but part of a Qualified Professional audit).
April 06, 2016 - IR27007 and Advisory for maximum daily flow exceedances, BOD exceedances and residual chlorine outside of required range.
Please submit all annual/quarterly/monthly reports and data submissions to the Ministry's Routine Environmental Reporting Submission Mailbox at EnvAuthorizationsReporting@gov.bc.ca"
More information about the reporting requirements may be found at www2.gov.bc.ca/gov/content/environment/waste-management/waste-discharge-authorization/data-and-report-submissions/routine-environmental-reporting-submission-mailbox"

Below are attachments related to this inspection.

Please be advised that this inspection report may be published on the provincial government website in 7 days.

If you have any questions about this letter, please contact the undersigned.

Yours truly,

Darren Stewart

Environmental Protection Officer

Attachments:
1) IR76276 Photo Record.pdf Photo Record

Deliver via:
Email: [X] **Fax:** [ ] **Mail:** [ ]
Registered Mail: [ ] **Hand Delivery:** [ ]

Ministry of Environment and Climate Change Strategy
Compliance
Environmental Protection Division
Mailing Address:
2080-A
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Nanaimo BC V9E 6J9
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Website: www.gov.bc.ca/env

DISCLAIMER:
Please note that sections of the permit, regulation or code of practice referenced in this inspection record are for guidance and are not the official version. Please refer to the original permit, regulation or code of practice.

To see the most up to date version of the regulations and codes of practices please visit
http://www.bclaws.ca

If you require a copy of the original permit, please contact the inspector noted on this inspection record.

It is also important to note that this inspection record does not necessarily reflect each requirement or condition of the authorization therefore compliance is noted only for the requirements or conditions listed in the inspection record.

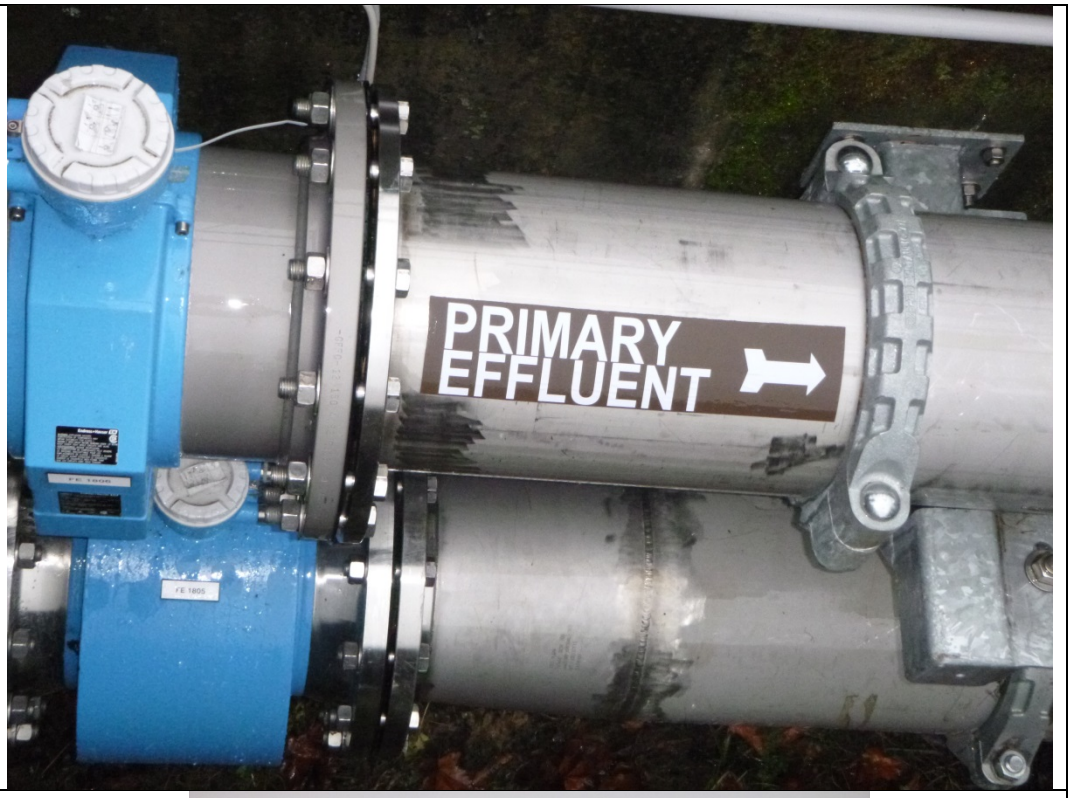
Authorization: 120	Client Name: Town of Ladysmith
NRIS IR #: 76276	Site Inspection Date: January 11, 2018

Photo 1 Influent Screening	
Photo 2 Two Salsnes Fliters used for primary treatment	

Authorization: 120	Client Name: Town of Ladysmith
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Photo 3 Secondary (biological) treatment using a moving bed bioreactor (MMBR) inside large tank. Inset: View inside the MMBR	
Photo 4. Dissolved air flotation (DAF) tanks and skimmers for clarification (separation and removal of biological solids).	

Authorization: 120	Client Name: Town of Ladysmith
NRIS IR #: 76276	Site Inspection Date: January 11, 2018

Photo 5 Endress + Hauser Flow meters. Lower pipe is the treated effluent discharge.	
Photo 6 Outfall signage in the water. Required signage on shore above the high water mark was not confirmed.	